SLOWING, COOLING, AND TRAPPING: TUTORIAL QUESTIONS

1- Magneto-optical trap

Consider a magneto-optical trap for Cs atoms. There is a number of allowed transitions between the $^2S_{1/2}$ ground state hyperfine levels and the hyperfine levels of the $^2P_{1/2}$ and $^2P_{3/2}$ excited state. Which transition would be best for the creation of the magneto-optical trap, and why?

2- The Scattering Force

Atomic beam of Cs (M=132 amu) is produced in a source at 300 K and slowed by counter-propagating laser radiation, using a transition with a wavelength of 852 nm and a lifetime of the excited state of 30 ns. What is the stopping distance at half the maximum deceleration?

How many photons are scattered to stop the atom?

Answer the above questions for a beam of hydrogen atoms cooled on the Lyman-α transition (121.6 nm), where the excited state lives for 1.6 ns.

$h = 6.62607004 \times 10^{-34}\, Js$
$k_B = 1.38064852 \times 10^{-23}\, J/K$
$\mathrm{M}_{Cs} = 2.20694688 \times 10^{-25}\, kg$

3- Lasers

Explain why it is difficult to construct lasers in the UV range.

4- Slowing and cooling with lasers

a. Consider an atom with a short excited-state lifetime (**A**) and an atom with a long excited-state lifetime (**B**), all other properties being the same. Which atom, **A** or **B**,
a) can be cooled at a higher rate using the scattering force?
b) can be cooled to a lower Doppler-limit temperature?
Explain your answer.

b. Consider an atom with a short excitation wavelength (**C**) and an atom with a long excitation wavelength (**D**), all other properties being the same.
Which atom, **C** or **D**, has the larger recoil velocity? Explain your answer.

c. What makes a transition suitable for laser cooling? Give at least 3 criteria.